

Science, Change, and Uncertainty Scientists have gathered lots of important information that helps cure and prevent disease, grow food, and link the world electronically. Yet much of nature remains a mystery. Almost every scientific discovery raises more questions than it answers. Often, research yields surprises that point future studies in new and unexpected directions. This constant change doesn't mean science has failed. On the contrary, it shows that science continues to advance.

That's why studying science means more than just memorizing what we know. It also means understanding what we don't know. You may be surprised to hear this, but science rarely "proves" anything. Scientists aim for the best understanding of the natural world that current methods can reveal. Uncertainty is always part of the scientific process, and is part of what makes science exciting!

We hope to show you that understanding science isn't just about learning "facts." We hope you'll gain some understanding of the spirit of scientific inquiry, of the way scientists think, and of both the process and excitement of discovery. Don't just memorize today's scientific facts and ideas. And please don't believe them, just because they are in a textbook! Instead, try to understand how scientists developed those ideas. Pose the kinds of questions scientists ask. Try to see the thinking behind the experiments we describe.

 **READING CHECK** **Construct an Explanation** How is scientific knowledge different from other types of knowledge?



INTERACTIVITY

Discover the power of scientific methodology.

Scientific Methodology

Is science a mysterious process that only scientists do under special circumstances? Nope! You use scientific thinking all the time! Suppose your family's car won't start. What do you do? You use what you know about cars to ask questions. Is the battery dead? You test that idea by turning the key in the ignition. If the starter motor works but the engine doesn't start, you reject the dead battery idea. Is the car out of gas? A glance at the fuel gauge tests that idea. Again and again, you apply scientific thinking until the problem is solved—or until you run out of ideas and call a mechanic!

Scientists work in pretty much the same way. There isn't a single, cut-and-dried "scientific method." But there is a general style of investigation we call scientific methodology, which is a fancy way of saying "the way science works." **Q Scientific methodology involves observing and asking questions, forming hypotheses, conducting controlled experiments, collecting and analyzing data, and drawing conclusions.** Figure 1-2 shows how one research team used scientific methodology in its study of one particular species of marsh grass in a New England salt marsh.

